Energy-Efficient Hybrid STAR-RIS via Sparse Active Amplification and Gain Control

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***Abstract*—Hybrid active-passive simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) improve spectral efficiency (SE), but their energy efficiency (EE) is governed by amplifier noise, active-set sparsity, and hardware power. This paper develops an equivalent post-ZF scalar-surrogate framework for approximating the achievable EE-SE frontier of hybrid STAR-RISs. A multi-component power model captures PA-normalized BS transmit power, gain-dependent amplification overhead, controller/bias power, and dynamic switching overhead. For a fixed sparse active support, we derive a closed-form SE-optimal common-gain law, its boundary cases, and a gain-on threshold showing when amplification offsets injected noise. We further refine the gain through a side-wise power surrogate with a sufficient convexity condition. A finite-grid ϵ-constraint solver with a sparse ranker pool produces reproducible frontier points under ES, TS, and SS protocols. Simulations show that sparse activation improves EE at moderate-to-high SE while avoiding the burden of side-assigned fully active operation.**

***Index Terms*—STAR-RIS, active RIS, hybrid active-passive RIS, energy efficiency, spectral efficiency, Pareto frontier, amplifier noise.**

I. Introduction

Simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) extend passive beam manipulation to simultaneous transmission and reflection, allowing a common surface to serve users on both sides of the surface [2], [3]. This capability is attractive for asymmetric indoor-outdoor and cell-edge coverage. However, purely passive STAR-RIS still suffers from multiplicative path loss [1].

Sparse active elements can mitigate this loss at the price of amplifier noise and hardware power [4]–[7]. Measurement-driven studies further show that RIS power contains controller/bias and switching terms rather than a single constant [8], [9]. In STAR-RIS, these effects become more pronounced because the two sides compete for aperture, power split, and active hardware budget.

Existing active or hybrid STAR-RIS studies have addressed important neighboring scenarios, including NOMA, SWIPT, URLLC, and ISAC [10]–[14]. Nevertheless, the joint role of amplifier noise, sparse active-set selection, dynamic switching overhead, and side-wise EE-SE tradeoff has not been characterized under a common structured frontier-generation framework. This paper therefore develops an equivalent post-ZF scalar surrogate for hybrid active-passive STAR-RIS and uses it to derive tractable, reproducible achievable-frontier approximations. The main contributions are as follows.

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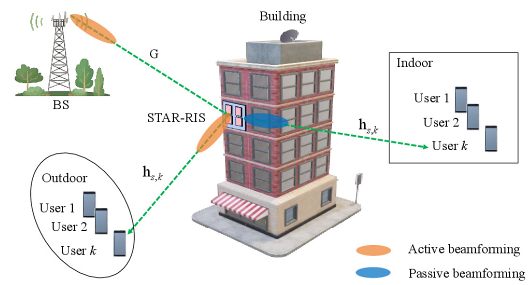

Fig. 1. Hybrid STAR-RIS-aided wireless system considered in this paper.

- We formulate a hybrid active-passive STAR-RIS downlink and explicitly distinguish the original full-dimensional MISO model from the equivalent post-ZF scalar surrogate used for structural analysis and simulations.
- We introduce a multi-component power model consisting of PA-normalized BS transmit power, gain-dependent amplification overhead, static controller/bias power, and dynamic switching overhead. We also state how a stricter signal-dependent active-output constraint can be incorporated.
- We derive a conservative side-wise SINR law, a closed-form SE-optimal common gain with boundary cases, a fixed-support gain-on threshold, and a power-minimizing scalar refinement with a sufficient convexity condition.
- We develop a finite-grid ϵ-constraint solver based on a cost-aware sparse ranker pool and report additional baselines, feasibility probabilities, runtime-complexity scaling, and power-breakdown results across ES, TS, and SS.

II. System model

A. Hybrid active-passive STAR-RIS downlink

Consider a BS with M antennas serving K_t users on the transmission side and K_r users on the reflection side via an N-element STAR-RIS. Let $\mathcal{K}_t = \{1, \ldots, K_t\}$ and $\mathcal{K}_r = \{1, \ldots, K_r\}$. Under the ES protocol [3], the nth STAR-RIS element uses transmission and reflection coefficients

$$\theta_{n,s} = \sqrt{\rho_{n,s}}\beta_{n,s} e^{j\phi_{n,s}}, \qquad s \in \{t, r\}, \tag{1}$$

with $\rho_{n,t} + \rho_{n,r} = 1$, $0 \leq \rho_{n,s} \leq 1$, and $1 \leq \beta_{n,s} \leq \beta_{\max}$, where passive elements use $\beta_{n,s} = 1$ and active elements use $\beta_{n,s} > 1$ on their assigned side. Let $\mathcal{A}_t$ and $\mathcal{A}_r$ denote the active-element sets for transmission and reflection, respectively. We assume side-assigned active RF chains, so $\mathcal{A}_t \cap \mathcal{A}_r = \emptyset$, and let $\mathcal{A} = \mathcal{A}_t \cup \mathcal{A}_r$.

Denote by $\mathbf{G} \in \mathbb{C}^{N\times M}$ the BS-to-STAR-RIS channel, and by $\mathbf{h}_{s,k} \in \mathbb{C}^N$ the STAR-RIS-to-user channel on side s for user k. The received signal of user (s,k) is

$$y_{s,k} = \mathbf{h}_{s,k}^H \boldsymbol{\Theta}_s \mathbf{G} \sum_{s'\in\{t,r\}} \sum_{j\in\mathcal{K}_{s'}} \mathbf{w}_{s',j} x_{s',j} + \mathbf{h}_{s,k}^H \mathbf{n}_{a,s} + n_{s,k}, \tag{2}$$

where $\boldsymbol{\Theta}_s = \mathrm{diag}(\theta_{1,s},\ldots,\theta_{N,s})$, $\mathbf{w}_{s,j} \in \mathbb{C}^M$ is the BS beamformer for user (s,j), $x_{s,j}$ is the unit-power information symbol, and $n_{s,k} \sim \mathcal{CN}(0,\sigma_0^2)$ is thermal noise. The vector $\mathbf{n}_{a,s}$ collects the amplifier noises of the active elements on side s, and we model it as [4], [7]

$$\mathbf{n}_{a,s} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_{a,s}), \qquad \mathbf{C}_{a,s} = \mathrm{diag}(c_{1,s},\ldots,c_{N,s}), \tag{3}$$

where $c_{n,s} = \rho_{n,s}\beta_{n,s}^2\sigma_a^2$ for $n \in \mathcal{A}_s$ and $c_{n,s} = 0$ otherwise. Here, σ_a^2 denotes the variance of the pre-amplification active-element noise, and $\mathbf{n}_{a,s}$ is the equivalent post-amplification noise vector. The SINR of user (s,k) becomes

$$\gamma_{s,k} = \frac{\left|\mathbf{h}_{s,k}^H \boldsymbol{\Theta}_s \mathbf{G} \mathbf{w}_{s,k}\right|^2}{\sum_{\substack{s'\in\{t,r\},\ j\in\mathcal{K}_{s'} \\ (s',j)\neq(s,k)}} \left|\mathbf{h}_{s,k}^H \boldsymbol{\Theta}_s \mathbf{G} \mathbf{w}_{s',j}\right|^2 + \sigma_{a,s,k}^2 + \sigma_0^2}, \tag{4}$$

where $\sigma_{a,s,k}^2 = \mathbf{h}_{s,k}^H \mathbf{C}_{a,s} \mathbf{h}_{s,k}$ is the active-noise term seen at user (s,k).

B. Power-consumption model

Following measurement-based RIS power models [8], [9], the consumed power is

$$\begin{aligned} P_{\mathrm{tot}} = &\frac{1}{\eta_{\mathrm{PA}}} \sum_{s\in\{t,r\}} \sum_{k\in\mathcal{K}_s} \|\mathbf{w}_{s,k}\|^2 + P_{\mathrm{ctrl}} + P_{\mathrm{bias}}|\mathcal{A}| \\ &+ f_u(E_{\mathrm{sw},0} + E_{\mathrm{sw},1}|\mathcal{A}|) \\ &+ \sum_{n\in\mathcal{A}_t} \mu_t(\beta_{n,t}^2 - 1) + \sum_{n\in\mathcal{A}_r} \mu_r(\beta_{n,r}^2 - 1), \end{aligned} \tag{5}$$

where the four terms are the PA-normalized BS transmit power, static controller/bias power, dynamic switching power, and gain-dependent amplification power. Since $E_{\mathrm{sw},0}$ and $E_{\mathrm{sw},1}$ are energies per update, $f_u(E_{\mathrm{sw},0} + E_{\mathrm{sw},1}|\mathcal{A}|)$ is measured in watts.

The term $\mu_s(\beta_{n,s}^2 - 1)$ in (5) models gain-dependent circuit/amplifier overhead rather than signal-dependent RF output power. A stricter active-output constraint based on the incident RF power can be incorporated in a full-dimensional beamforming model, but it is not part of the present scalar surrogate.

C. Equivalent scalar channel model

To approximate the EE-SE Pareto frontier with a tractable low-dimensional search, we use an idealized scalar model induced by post-ZF directions. Specifically, residual multiuser leakage in (4) is neglected, common side-wise splitting and gain are used with $\rho_{n,s} = \rho_s$ and $\beta_{n,s} = \beta_s$ for $n \in \mathcal{A}_s$, dominant cascaded paths are phase-aligned, and each side target is evenly split among its users.

Let g_n denote the equivalent post-ZF BS-to-element gain of element n. With $h_{s,k,n}$ denoting the nth entry of $\mathbf{h}_{s,k}$, define

$$u_{s,k,n} = |g_n|\,|h_{s,k,n}|. \tag{6}$$

Then the side-wise effective coherent term for user (s,k) is

$$\psi_{s,k}(\beta_s) = \sum_{n\in\mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n} + \beta_s \sum_{n\in\mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \tag{7}$$

where $\mathcal{P}_s = \{1,\ldots,N\} \setminus \mathcal{A}_s$ contains passive elements and those active only for the opposite side. The corresponding active-noise term is

$$\zeta_{s,k}(\beta_s) = \sigma_0^2 + \rho_s\beta_s^2\sigma_a^2 \sum_{n\in\mathcal{A}_s} |h_{s,k,n}|^2. \tag{8}$$

With equal per-user rate splitting, the side power required to meet target R_s is

$$p_s^{\min}(\beta_s) = \sum_{k\in\mathcal{K}_s} \frac{(2^{R_s/K_s} - 1)\zeta_{s,k}(\beta_s)}{\psi_{s,k}^2(\beta_s)}. \tag{9}$$

III. Problem formulation

The sum spectral efficiency is

$$R_\Sigma = \sum_{s\in\{t,r\}} \sum_{k\in\mathcal{K}_s} \log_2(1 + \gamma_{s,k}), \tag{10}$$

and the normalized system energy efficiency is

$$\mathrm{EE}_{\mathrm{norm}} = \frac{R_\Sigma}{P_{\mathrm{tot}}} \quad \text{(bit/J/Hz)}. \tag{11}$$

For bandwidth B, the corresponding physical energy efficiency is $BR_\Sigma/P_{\mathrm{tot}}$ in bit/J. The numerical study uses the normalized value in (11); the original normalized-EE-maximization problem can be written as

$$\begin{aligned} \max_{\mathbf{W},\,\boldsymbol{\rho},\,\boldsymbol{\phi},\,\boldsymbol{\beta},\,\mathcal{A}} \quad & \frac{R_\Sigma}{P_{\mathrm{tot}}} \\ \text{s.t.} \quad & \log_2(1+\gamma_{s,k}) \geq \bar{R}_{s,k},\ \forall s,k, \\ & \sum_{s,k} \|\mathbf{w}_{s,k}\|^2 \leq P_{\max}^{\mathrm{BS}}, \\ & 1 \leq \beta_{n,s} \leq \beta_{\max},\ \forall n \in \mathcal{A}_s, \\ & \beta_{n,s} = 1,\ \forall n \notin \mathcal{A}_s, \\ & \mathcal{A}_t \cap \mathcal{A}_r = \emptyset, \quad \mathcal{A} = \mathcal{A}_t \cup \mathcal{A}_r, \\ & \rho_{n,t} + \rho_{n,r} = 1,\ 0 \leq \rho_{n,s} \leq 1, \\ & \rho_{n,s}\beta_{n,s}^2\sigma_a^2 \leq \Gamma_{n,s},\ \forall n \in \mathcal{A}_s,\ s \in \{t,r\}, \end{aligned} \tag{12}$$

where $\Gamma_{n,s}$ is a maximum noise-injection or hardware-safety limit. Problem (12) is highly non-convex. The fractional objective and the coupling among beamforming, STAR coefficients, active gains, and active-set selection make the problem difficult to solve directly.

To trace an EE-SE Pareto-frontier approximation, we adopt the ϵ-constraint formulation

$$\begin{aligned} \mathcal{P}(\epsilon): \quad \min_{\mathbf{W},\,\boldsymbol{\rho},\,\boldsymbol{\phi},\,\boldsymbol{\beta},\,\mathcal{A}} \quad & P_{\mathrm{tot}} \\ \text{s.t.} \quad & R_\Sigma \geq \epsilon, \\ & \text{constraints of (12)}. \end{aligned} \tag{13}$$

For each feasible ϵ, a globally optimal solution of (13) yields a weakly Pareto-optimal point of the EE-SE tradeoff. Because the solver below uses a structured sparse-activation surrogate and finite grids, the generated curve should be interpreted as an achievable Pareto-frontier approximation rather than a certified global boundary. We next exploit the model structure to derive transferable design rules.

IV. Gain structure

A. Amplification on–off criterion

Given a side support, the first question is whether the common active gain should be switched on, i.e., whether increasing β_s above one improves the side-wise SINR surrogate. Define

$$\begin{aligned} a_s &= \min_k \sum_{n\in\mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n}, & b_s &= \min_k \sum_{n\in\mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \\ c_s &= \sigma_0^2, & d_s &= \max_k \rho_s \sigma_a^2 \sum_{n\in\mathcal{A}_s} |h_{s,k,n}|^2. \end{aligned} \tag{14}$$

Here, a_s and b_s are conservative worst-user coherent-gain aggregates from passive and active contributions, while c_s and d_s are the thermal-noise baseline and active-noise coefficient. With an auxiliary normalized side power $\bar{p}_s$, the worst-user SINR surrogate is

$$\underline{\gamma}_s(\beta_s) = \frac{\bar{p}_s(a_s + b_s\beta_s)^2}{c_s + d_s\beta_s^2}. \tag{15}$$

The coherent term grows linearly in β_s, whereas the injected active noise grows quadratically. Differentiating (15) gives

$$\frac{\mathrm{d}\underline{\gamma}_s}{\mathrm{d}\beta_s} = \frac{2\bar{p}_s(a_s + b_s\beta_s)(b_s c_s - a_s d_s \beta_s)}{(c_s + d_s\beta_s^2)^2}. \tag{16}$$

Thus, at the passive-gain point $\beta_s = 1$, amplification is locally useful if

$$\left.\frac{\mathrm{d}\underline{\gamma}_s}{\mathrm{d}\beta_s}\right|_{\beta_s=1} > 0 \iff b_s c_s > a_s d_s. \tag{17}$$

Equation (17) is the gain-on criterion: the active coherent-gain aggregate, weighted by the thermal-noise baseline, must dominate the passive coherent-gain aggregate weighted by the active-noise coefficient. After bias, switching, and support-induced changes in d_s are included, practical activation becomes more selective, especially when c_s is small relative to d_s.

Proposition 1. *For the regular hybrid active-passive case $a_s > 0$, $b_s > 0$, $c_s > 0$, and $d_s > 0$, $\underline{\gamma}_s(\beta_s)$ is unimodal on $[1, \beta_{\max}]$. Its SE-oriented gain is*

$$\beta_{s,\mathrm{SE}}^\star = \left[\frac{b_s c_s}{a_s d_s}\right]_{[1,\beta_{\max}]}, \tag{18}$$

where $[x]_{[l,u]} = \min\{u, \max\{l, x\}\}$. If $b_s = 0$, set $\beta_{s,\mathrm{SE}}^\star = 1$; if $a_s = 0$ with $b_s > 0$, or if $d_s = 0$ with $b_s > 0$, $\underline{\gamma}_s(\beta_s)$ is nondecreasing and $\beta_{s,\mathrm{SE}}^\star = \beta_{\max}$. If $|\mathcal{A}_s| = 0$, set $\beta_s = 1$.

Proof: In the regular case, the sign of (16) is determined only by $b_s c_s - a_s d_s \beta_s$, which changes sign once. Hence the SINR surrogate is unimodal and the stationary point is $b_s c_s/(a_s d_s)$, clipped to $[1, \beta_{\max}]$. The boundary cases follow directly from (16): $b_s = 0$ makes the derivative non-positive, while $a_s = 0$ with $b_s > 0$ or $d_s = 0$ with $b_s > 0$ makes the surrogate nondecreasing. ■

B. Power-minimizing gain

The gain in (18) is SE-oriented. EE optimization must also include BS transmit power and active-amplifier power. For a target side rate R_s, define the side-wise surrogate cost $\Psi_s(\beta_s)$ as

TABLE I
Case classification for the optimization of β_s

Case	Condition	Conclusion
Fully passive	$\lvert\mathcal{A}_s\rvert = 0$, $b_s = d_s = 0$	No active gain is to be optimized; set $\beta_s^\star = 1$.
Active	$\lvert\mathcal{A}_s\rvert > 0$, $a_s > 0$, $b_s = 0$	$\Psi_s'(\beta_s) > 0$; Ψ_s is strictly increasing; set $\beta_s^\star = 1$.
Fully active	$\lvert\mathcal{A}_s\rvert > 0$, $a_s = 0$, $b_s > 0$	$\Psi_s''(\beta_s) > 0$; Ψ_s is strictly convex and admits a unique minimizer.
Hybrid active-passive	$a_s > 0$, $b_s > 0$, $d_s > 0$	(22) ensures strict convexity on $[1, \beta_{\max}]$ and a unique minimizer.

$$\Psi_s(\beta_s) = \frac{K_s(2^{R_s/K_s} - 1)(c_s + d_s\beta_s^2)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^2} + \mu_s|\mathcal{A}_s|(\beta_s^2 - 1), \tag{19}$$

$$\Psi_s'(\beta_s) = \frac{2K_s(2^{R_s/K_s} - 1)(a_s d_s \beta_s - b_s c_s)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^3} + 2\mu_s|\mathcal{A}_s|\beta_s, \tag{20}$$

$$\begin{aligned}\Psi_s''(\beta_s) = {} & \frac{2K_s(2^{R_s/K_s} - 1)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^4}\Big(a_s^2 d_s - 2a_s b_s d_s \beta_s + 3b_s^2 c_s\Big) \\ & + 2\mu_s|\mathcal{A}_s|. \end{aligned} \tag{21}$$

Proposition 2. *Assume the hybrid active-passive case $a_s > 0$, $b_s > 0$, and $d_s > 0$. Under the condition*

$$\beta_{\max} \le \frac{a_s^2 d_s + 3b_s^2 c_s}{2a_s b_s d_s}, \tag{22}$$

$\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\max}]$. Consequently, the side-wise power-minimizing gain is unique and is obtained by checking the two endpoints and the interior stationary point solving (20), if it lies in $[1, \beta_{\max}]$.

Proof: Under (22), the bracketed term in (21) is non-negative on $[1, \beta_{\max}]$. Since $|\mathcal{A}_s| > 0$ and $\mu_s > 0$, the term $2\mu_s|\mathcal{A}_s|$ yields $\Psi_s''(\beta_s) > 0$ over the interval. Hence Ψ_s is strictly convex and has a unique minimizer. ■

Condition (22) is sufficient but not necessary. When it is violated, the bounded one-dimensional minimization is still used numerically. In the simulations, $\beta_s = 1$ is used in the first two cases. In the fully active and hybrid active-passive cases, and in any $d_s = 0$ active case, we minimize $\Psi_s(\beta_s)$ over $[1, \beta_{\max}]$; (18) is used only as an SE-oriented reference or initialization.

C. Extension to TS and SS

Since ES is covered by (14), for $p \in \{\mathrm{TS}, \mathrm{SS}\}$ let $\mathcal{P}_s^{(p)}$ and $\mathcal{A}_s^{(p)}$ denote the passive and active support sets. Define

$$\begin{aligned} a_s^{(p)} &= \min_k \sum_{n\in\mathcal{P}_s^{(p)}} u_{s,k,n}, & b_s^{(p)} &= \min_k \sum_{n\in\mathcal{A}_s^{(p)}} u_{s,k,n}, \\ c_s^{(p)} &= \sigma_0^2, & d_s^{(p)} &= \max_k \sigma_a^2 \sum_{n\in\mathcal{A}_s^{(p)}} |h_{s,k,n}|^2. \end{aligned}$$

For TS, $\mathcal{P}_s^{(\mathrm{TS})} = \{1,\ldots,N\} \setminus \mathcal{A}_s$ and the effective target becomes R_s/τ_s. Its consumed power is time-averaged:

$$P_{\mathrm{tot}}^{\mathrm{TS}} = P_{\mathrm{ctrl}} + P_{\mathrm{bias}}|\mathcal{A}| + f_u(E_{\mathrm{sw},0} + E_{\mathrm{sw},1}|\mathcal{A}|) + \sum_{s\in\{t,r\}} \tau_s \left(\frac{P_{\mathrm{tx},s}}{\eta_{\mathrm{PA}}} + \mu_s|\mathcal{A}_s|(\beta_s^2 - 1) \right), \tag{23}$$

where $P_{\mathrm{tx},s}$ is the BS transmit power in slot s. For SS, $\mathcal{P}_s^{(\mathrm{SS})} = \mathcal{S}_s \setminus \mathcal{A}_s$, with $\mathcal{S}_t \cup \mathcal{S}_r = \{1,\ldots,N\}$ and $\mathcal{S}_t \cap \mathcal{S}_r = \emptyset$. The ES rules then apply after replacing (a_s, b_s, c_s, d_s) by $(a_s^{(p)}, b_s^{(p)}, c_s^{(p)}, d_s^{(p)})$ and using the corresponding consumed-power denominator.

V. Sparse activation algorithm

For a target sum-SE level ϵ, the solver follows the execution order in Algorithm 1: it scans sparsity and rate-split grids, constructs a small pool of side-assigned active supports, updates the side-wise gains, and keeps the feasible candidate with the minimum consumed power.

A. Grid search over sparsity and rate split

The outer grid contains the active-cardinality budget $L \in \mathcal{L}$ and the side-rate split $\alpha \in \mathcal{G}_\alpha$. For each pair (L, α), the side targets are

$$R_t = \alpha\epsilon, \qquad R_r = (1-\alpha)\epsilon.$$

Thus, L controls the hardware sparsity, while α controls the traffic asymmetry and the side-demand weights used by the active-set rankers. Each grid point defines one candidate design subproblem: choose L side-assigned active elements, update (β_t, β_r), evaluate the scalar-surrogate constraints, and compare the resulting P_{tot}.

B. Active-set construction

For a fixed (L, α), we build candidate supports by testing whether an inactive element $n \notin \mathcal{A}_s$ should be switched on for side s. The test is based on the first-order change of the side-wise surrogate cost. From (19), define

$$C_s = \frac{K_s(2^{R_s/K_s} - 1)}{\eta_{\mathrm{PA}}},$$

which collects the side-rate and PA-efficiency factors in the transmit-power term. Introduce a relaxed activation variable $x_{n,s} \in [0,1]$, with $x_{n,s} = 0$ denoting inactive and $x_{n,s} = 1$ denoting active. Switching on element n changes its coefficient from passive gain 1 to active gain β_s, giving the worst-user coherent increment

$$\Delta u_{n,s} = \sqrt{\rho_s}|g_n| \min_k |h_{s,k,n}|.$$

The coherent denominator is therefore perturbed as

$$D_{n,s}(x_{n,s}) = a_s + b_s\beta_s + (\beta_s - 1)x_{n,s}\Delta u_{n,s}.$$

The active-noise coefficient changes from d_s to $d_s + x_{n,s}\Delta d_{n,s}$. Since the exact increment depends on the maximizing user after insertion, the ranking step uses

$$\Delta d_{n,s} \approx \rho_s\sigma_a^2\overline{|h_{s,k,n}|^2}, \qquad \overline{|h_{s,k,n}|^2} \triangleq \frac{1}{K_s}\sum_{k\in\mathcal{K}_s}|h_{s,k,n}|^2.$$

The resulting local cost perturbation is

$$\Psi_s(x_{n,s}) \approx C_s \frac{c_s + \beta_s^2(d_s + x_{n,s}\Delta d_{n,s})}{D_{n,s}^2(x_{n,s})} + \mu_s(|\mathcal{A}_s| + x_{n,s})(\beta_s^2 - 1) + x_{n,s}(P_{\mathrm{bias}} + f_u E_{\mathrm{sw},1}).$$

Differentiating at $x_{n,s} = 0$ gives

$$\left.\frac{\partial\Psi_s}{\partial x_{n,s}}\right|_{x_{n,s}=0} = -\kappa_{1,s}\Delta u_{n,s} + \kappa_{2,s}\Delta d_{n,s} + P_{\mathrm{bias}} + f_u E_{\mathrm{sw},1} + \mu_s(\beta_s^2 - 1),$$

where

$$\kappa_{1,s} = \frac{2C_s(c_s + d_s\beta_s^2)(\beta_s - 1)}{(a_s + b_s\beta_s)^3}, \qquad \kappa_{2,s} = \frac{C_s\beta_s^2}{(a_s + b_s\beta_s)^2}.$$

Hence, element n is beneficial in the first-order sense when

$$\tilde{\xi}_{n,s} \triangleq \frac{\kappa_{1,s}\Delta u_{n,s}}{\kappa_{2,s}\Delta d_{n,s} + P_{\mathrm{bias}} + f_u E_{\mathrm{sw},1} + \mu_s(\beta_s^2 - 1)} > 1.$$

This condition says that the transmit-power saving from coherent-gain improvement must exceed the active-noise, bias/switching, and amplifier-power penalties.

For low-complexity ranking, we use the normalized score

$$\xi_{n,s} = \frac{\omega_s\sqrt{\rho_s}|g_n| \min_k |h_{s,k,n}|}{\lambda_1\rho_s\sigma_a^2\overline{|h_{s,k,n}|^2} + \lambda_2(P_{\mathrm{bias}} + f_u E_{\mathrm{sw},1})}, \tag{24}$$

where $(\omega_t, \omega_r) = (\alpha, 1-\alpha)$ and λ_1, λ_2 are normalization weights. In the simulations, $\lambda_1 = \sigma_0^{-2}$ and $\lambda_2 = (P_{\max}^{\mathrm{BS}})^{-1}$. The score in (24) is used only to generate candidate supports: for each element, the larger side-wise score determines its assigned side, and the top L elements form one candidate support. A strength-only ranker based on $\omega_s|g_n| \min_k |h_{s,k,n}|$ forms a second candidate support and is also reported as the "strongest greedy" baseline. All candidates are re-evaluated by (9), (19), and (5) after gain update.

C. Gain update

For every candidate $(\mathcal{A}_t, \mathcal{A}_r)$, the solver computes (a_s, b_s, c_s, d_s) for $s \in \{t, r\}$ and updates the common gain on each side. If $|\mathcal{A}_s| = 0$ or $b_s = 0$, no useful active coherent term is available and we set $\beta_s = 1$. Otherwise, we solve

$$\min_{\beta_s\in[1,\beta_{\max}]} \Psi_s(\beta_s).$$

Proposition 2 gives uniqueness under its sufficient condition; (18) is used only as an SE-oriented reference or initialization. The updated gains determine the required BS power through (9). The candidate is feasible if it satisfies the BS power budget, gain cap, disjoint side assignment, protocol-specific ES/TS/SS constraints, and the optional safety limit $\Gamma_{n,s}$ within the scalar surrogate. The feasible candidate with the smallest P_{tot} is retained.

D. Complexity

Algorithm 1 terminates because $\mathcal{L}$ and $\mathcal{G}_\alpha$ are finite. For one boundary point, passive and fixed-uniform designs require $\mathcal{O}(G(NK + Q_\beta))$ operations, where $G = |\mathcal{G}_\alpha|$, $K = K_t + K_r$, and Q_β is the one-dimensional gain-search budget. Scanning the active-cardinality grid adds $L_g = |\mathcal{L}|$. The proposed ranker pool also sorts element scores, adding an $N\log N$ term; hence its dominant per-boundary-point complexity is

$$\mathcal{O}\big(L_g G(N\log N + NK + Q_\beta)\big).$$

Algorithm 1 Sparse-Activation Solver

Require: Target ϵ, grids $\mathcal{L}$, $\mathcal{G}_\alpha$, gain cap $\beta_{\max}$

$P_{\min} \leftarrow +\infty$, best $\leftarrow \emptyset$
for $L \in \mathcal{L}$, $\alpha \in \mathcal{G}_\alpha$ **do**
  Generate support pool using (24) and strength-only ranking
  for all $(\mathcal{A}_t, \mathcal{A}_r)$ in the pool **do**
    Update side coefficients and gains
    Evaluate required BS power and P_{tot}
    if all scalar-surrogate constraints hold and $P_{\text{tot}} < P_{\min}$ **then**
      $P_{\min} \leftarrow P_{\text{tot}}$, best $\leftarrow (\mathcal{A}_t, \mathcal{A}_r, \beta_t, \beta_r, \alpha)$
    end if
  end for
end for
return best

The single-random sparse baseline has the same order multiplied by Q_{rand}; the reported setting uses $Q_{\text{rand}} = 1$.

VI. Numerical Results

We consider $N = 24$, $K_t = K_r = 2$, Nakagami-m fading, and the normalized parameters in Table II. They validate the structured surrogate rather than a calibrated carrier-frequency deployment. In the simulations, g_n is generated directly as an equivalent post-ZF element gain, not extracted from an explicit M-antenna channel realization. Each curve averages 100 independent seeds. Unless stated otherwise, $P_{\max}^{\text{BS}} = 5$ W, $(\rho_t, \rho_r) = (0.55, 0.45)$, $(\tau_t, \tau_r) = (0.55, 0.45)$, and $\beta_{\max} = 5$. The safety limit is inactive.

Reported EE values are per-realization averages, $\mathbb{E}[R_\Sigma / P_{\text{tot}}]$, whereas power-breakdown terms are averaged separately; hence they should not be interpreted as $R_\Sigma / \mathbb{E}[P_{\text{tot}}]$.

TABLE II
Simulation Parameters

Parameter	Value
STAR-RIS elements N	24
Users (K_t, K_r)	$(2, 2)$
Fading model	Nakagami-m
Nakagami parameters (m_{BR}, m_{RU})	$(1.0, 1.0)$
Avg. powers/path losses $(\Omega_{BR}, \Omega_t, \Omega_r)$	0.18, $(0.064, 0.052)$, $(0.080, 0.064)$
ES coefficients (ρ_t, ρ_r)	$(0.55, 0.45)$
TS slot fractions (τ_t, τ_r)	$(0.55, 0.45)$
Noise std. (σ_0, σ_a)	$(0.26, 0.05)$
Maximum active gain $\beta_{\max}$	5
BS power budget $P_{\max}^{\text{BS}}$	5.0 W
Static controller power P_{ctrl}	0.12 W
Bias power P_{bias}	0.016 W per element
Update frequency f_u	50 Hz
Switching energy/update $(E_{\text{sw},0}, E_{\text{sw},1})$	$(2.0, 1.2) \times 10^{-4}$ J/update
Amplifier coefficient $\mu_t = \mu_r$	0.0025 W
PA efficiency η_{PA}	0.38
Monte Carlo trials / seeds	$100 / 1, \ldots, 100$
Rate-split grid $\mathcal{G}_\alpha$	$\{0.20, 0.25, \ldots, 0.80\}$
Active-cardinality grid $\mathcal{L}$	$\{0, 3, \ldots, 24\}$; fixed uniform uses $L = 6$
Single-random repetitions Q_{rand}	1
Safety limit $\Gamma_{n,s}$	inactive in simulations

The ES baselines are: passive operation; fixed uniform activation with $L = 6$; optimized uniform activation over $L \in \mathcal{L}$; a lightweight single-random sparse support per (L, α); a strongest-greedy ablation; side-assigned fully active operation with $L = N$; and the proposed two-ranker pool followed by surrogate power re-evaluation.

Passive feasibility drops to 0.94 and 0.67 at $R_\Sigma = 18$ and 20 bit/s/Hz, respectively; all active/hybrid baselines remain feasible in these selected points.

Fig. 2 shows the ES frontier. Passive STAR-RIS is competitive at low SE but becomes partly infeasible at high SE. The side-assigned fully active benchmark lowers BS transmit power but pays large hardware overhead. At $R_\Sigma = 12$ bit/s/Hz, the proposed design achieves EE 8.67 bit/J/Hz, improving over fixed uniform, optimized uniform, single-random sparse, side-assigned active, and passive operation by 33.3%, 19.7%, 15.3%, 31.7%, and 98.8%, respectively; it also slightly exceeds the strongest-greedy ablation.

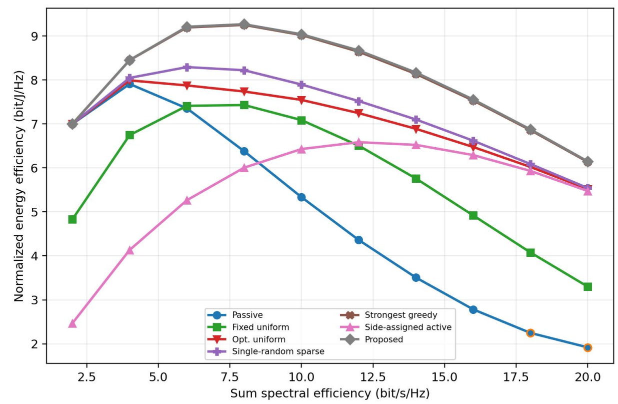

Fig. 2. Approximated ES EE-SE frontier. Hollow markers indicate conditional averages with feasibility probability below one.

Fig. 3 compares ES, TS, and SS. TS requires the densest active set because slot targets inflate by $1/\tau_s$. At $R_\Sigma = 12$ bit/s/Hz, the active ratios are about 0.30, 0.54, and 0.32, with average gains 4.68, 4.96, and 4.65, respectively.

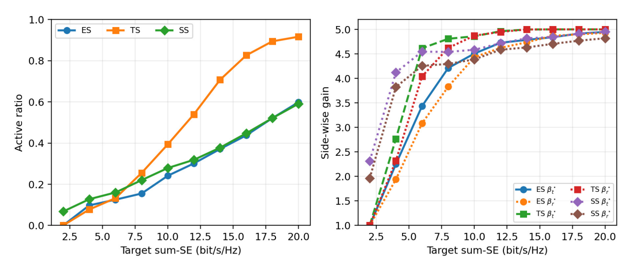

Fig. 3. Cross-mode design rules across ES, TS, and SS: active ratio and side-wise common gain.

Fig. 4 shows hardware sensitivity at $R_\Sigma = 12$ bit/s/Hz. The gain decreases with σ_a, most sharply under TS. Raising the update frequency from 50 Hz to 550 Hz reduces EE for all protocols, with TS penalized most by its denser activation.

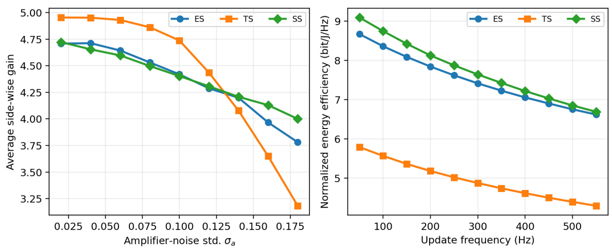

Fig. 4. Hardware sensitivity across ES, TS, and SS: average side-wise gain versus amplifier noise and EE versus switching overhead.

Fig. 5 compares ES power consumption at $R_\Sigma = 12$ and 16 bit/s/Hz. At $R_\Sigma = 12$ bit/s/Hz, side-assigned fully active operation has the lowest PA-normalized BS transmit power, about 0.70 W, but adds about 0.66 W static/dynamic overhead and 0.49 W amplification power. Passive operation needs about 2.78 W BS power, whereas the proposed sparse design gives the lowest total power, about 1.41 W. The same trend persists at $R_\Sigma = 16$ bit/s/Hz: sparse activation avoids both the BS-power burden of passive operation and the hardware-power burden of dense active operation.

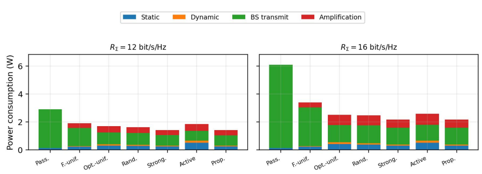

Fig. 5. ES power breakdown at $R_\Sigma = 12$ and 16 bit/s/Hz.

VII. Conclusion

This paper studied the EE-SE tradeoff of hybrid active-passive STAR-RIS through an equivalent post-ZF scalar surrogate. The analysis linked amplifier noise, active-set sparsity, side-wise asymmetry, and static/dynamic hardware power. For fixed support, we derived an SE-optimal gain law, a gain-on threshold, and a power-minimizing refinement with a sufficient convexity condition. The finite-grid solver then generated reproducible achievable-frontier points. Results show that passive STAR-RIS is attractive at low SE, whereas sparse hybrid activation yields higher EE at moderate-to-high SE without full-array active overhead. Future work should add full-dimensional beamforming verification, signal-dependent output constraints, CSI uncertainty, and calibration-aware protocol design.

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